Gaurav Kuwar

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Summary

- I work on cryptographic hardware accelerator research (zero-knowledge proofs (ZKP), elliptic-curve and modular arithmetic) on both the architecture and electronic design automation (EDA) sides, with a first-author publication. At the same time I work full-time as a software engineer, with prior industry experience spanning security and AI/ML.

Education

- **New York University** | *Master of Science - Computer Engineering* *Jan 2026*
- **City College of New York** | *Bachelor of Science - Computer Science* *Jun 2023*

Publications

- **Locus: A Framework for Exploring and Optimizing Point Addition Hardware for Zero-Knowledge Proofs**: **Gaurav Kuwar**, Alhad Daftardar, Jianqiao Mo, Siddharth Garg, Brandon Reagen. IEEE/ACM International Conference on Computer-Aided Design (ICCAD), November 2026.

Talks

- **Cryptolets: An Open-Source Hardware Framework for Cryptographic Kernels**: **Gaurav Kuwar**, Brandon Reagen. Demo and poster, Midscale Experimental Research Infrastructure Forum (MERIF), Washington, DC, September 2026.
- **Tessera: An Open-Source Hardware Framework for Cryptographic Kernels**: **Gaurav Kuwar**, Alhad Daftardar, Jianqiao Mo, Brandon Reagen. Workshop on Open-Source Computer Architecture Research (OSCAR) at the International Symposium on Computer Architecture (ISCA), June 2026.
- **Tessera Demo**: *Cryptolets: Building a Community and Open-Source Hardware Repository for Cryptographic Computing*, tutorial at the International Symposium on Computer Architecture (ISCA), June 2026.

Experience

- **New York University** | *Graduate Research Assistant* *Brooklyn, NY | May 2025 – Present*
  - Led Locus, a High-Level Synthesis (HLS) based framework that generates optimized elliptic-curve point addition hardware for ASIC and FPGA; explored 1,000+ design points and achieved 2.71× geomean speedup and 3.11× area reduction over prior ASICs. (ICCAD 2026)
  - Generalizing Locus into Tessera, an HLS framework and kernel library covering large-integer, modular, and elliptic-curve arithmetic for privacy-preserving computation; built a Python reference cryptographic library to verify the C++ HLS source and generated RTL. (OSCAR @ ISCA 2026)
  - Designing a programmable Multi-Scalar Multiplication (MSM) accelerator for ZKPs that runs multiple MSM algorithms on a shared datapath, selected per workload at runtime.
  - Researching LLMs and agentic AI for HLS-based chip design of higher-level cryptographic kernels.
- **Capital One** | *Software Engineer* *Washington, D.C. | Feb 2026 – Present*
  - Within the Bank Marketing Technology org, part of a new, fast-paced team which owns automated customer payout execution for customer offers; designing and driving changes for a new initiative through presentation, discussion, and implementation for system design choices.
  - Improved the service's testing posture, adding coverage across unit, component, and live-dependency layers.
  - Integrated multi-agent AI workflows with Claude Code across research, design, implementation, and technical writing, accelerating delivery by turning written specs into reviewed, production-ready code.
- **SeatGeek** | *Software Engineer Intern* *New York, NY | Jun 2025 – Aug 2025*
  - Built SeatGeek's first internal bot-classification baseline from scratch, spanning data collection, cleaning, feature engineering, and model experimentation.
  - Experimented with supervised, unsupervised, and Louvain community detection on graphs.
- **NVIDIA** | *Software Engineer Intern* *Santa Clara, CA | Sep 2024 – Dec 2024*
  - Built a scalable OAuth 2.0-based authentication proxy with FastAPI and NVIDIA's identity federation, enabling Single Sign-On and service-to-service auth with token validation, access control, and async request proxying.
  - Proposed and deployed a proxy-based architecture that integrated cleanly into the security portal, enhancing security, reducing latency, and requiring minimal changes to downstream services.
- **Capital One** | *Software Engineer Intern* *McLean, VA | Jun 2024 – Aug 2024*
  - Architected and deployed a full-stack internal platform using Flask, Angular, and AWS Fargate to eliminate manual data registration delays, cut turnaround from 2 days to 30 minutes, and improve team efficiency.
- **Yahoo** | *Software Engineer Intern* *New York, NY | Jun 2023 – Aug 2023*
  - Automated security compliance reporting with Python-based extraction from AWS DynamoDB and Screwdriver CI/CD workflows, cutting baseline release time 5x; explored container security using Kubernetes Goat.

Projects

- **Cryptolets**: Open-source initiative building a shared hardware repository for cryptographic computing, lowering the barrier to entry for privacy-preserving hardware research; developed under the $3.8M NSF CIRC GRAND award at NYU. Tessera is the core hardware framework in Cryptolets; Tessera first began as Locus. (cryptolets.org, github.com/cryptolets/cryptolets)
- **Programmable MSM Accelerator for Zero-Knowledge Proofs**: Designed an ASIC architecture for multi-scalar multiplication, the dominant kernel in ZKP proof generation. A shared datapath runs multiple MSM algorithms, with algorithm and window size selected per workload at runtime. Built a cycle-level simulator to model the architecture and evaluate it against prior ASIC designs.
- **ML Inference Accelerator on FPGA**: Built a custom hardware accelerator for quantized neural network inference in VHDL, with a pipelined MAC array, on-chip activation memory, and FSM-based dataflow, achieving sub-100μs end-to-end inference latency at 50 MHz. Wrote every component from scratch, including the multipliers, which are emitted by a custom parameterizable Python-to-VHDL RTL generator. (github.com/gauravkuwar/array-mul-gen)
- **Jailbreaking Deep Models**: Implemented adversarial attacks (FGSM, PGD, patch-based) on ResNet-34 using PyTorch and a subset of ImageNet; degraded top-1 accuracy from 76% to 0% under the strongest attack and evaluated transferability to DenseNet-121. Tech: Python, PyTorch (github.com/gauravkuwar/deep-learning-proj-3)

Skills

- **Hardware**: HLS (Catapult, Vitis), RTL (Verilog, VHDL), ASIC/FPGA flows, Design Compiler, Vivado, ModelSim
- **Languages**: C++, Python, C, TypeScript
- **Software & ML**: PyTorch, FastAPI, Docker, AWS, Agentic AI development (Claude Code, Codex)